Komal Chandiramani

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komalchandiramani | komal-ml

Technical Skills

Programming Languages	Python, SQL, Java, C++
Machine Learning	PyTorch, NetworkX, Scikit-Learn, W&B
Cloud & Infrastructure	AWS, Docker, PostgreSQL, MongoDB
Data Engineering & Big Data	Airflow, Apache Spark, Hadoop

Work Experience

National University of Singapore — Jan 2026 – Apr 2026
Research Assistant – Machine Learning Specialist — Singapore

- Built a data cleaning and processing pipeline for a large-scale supermarket dataset. Used text embeddings and graph networks to aggregate similar products, reducing product dimensionality by ∼60%.
- Implemented a custom utility estimation model in PyTorch and JAX, translating academic research specifications into a functional and scalable codebase.
- Set up the training pipeline using JIT-compiled batched gradient ascent in JAX/Optax, with Adam, learning rate scheduling, and Weights & Biases tracking for experiment reproducibility.
- Implemented the Metropolis-Hastings algorithm to generate simulated datasets, used to evaluate how well the model captures product purchasing patterns.

Asian Institute of Digital Finance — Jan 2025 – Apr 2025
Research Assistant – Data Engineer — Singapore

- Architected an ETL pipeline using Airflow and Docker to extract and process 1,000+ SEC filings monthly, streamlining data availability for downstream financial analysis.
- Built CI/CD pipelines (GitLab) to support reliable deployment and testing of data workflows.

Blue Lion Labs — Sep 2022 – Oct 2024
Machine Learning Engineer — Ontario, Canada

- Developed a pipeline for 10,000+ images that automated data preprocessing and data splitting, ensuring balanced datasets for reliable Harmful Algal Bloom detection.
- Trained and optimized deep learning models (YOLOv5) for object detection and segmentation.
- Conducted experiments for optimization and implemented experiment tracking using Weights & Biases.
- Presented weekly visual performance reports using Python and Excel to communicate technical model results to cross-functional teams, ensuring project transparency.

ContextLogic – Wish — Jan 2022 – Apr 2022
Data Science Intern — Ontario, Canada

- Designed a Tableau dashboard to monitor KPIs, translating complex data into key insights to support business acumen and product strategy.
- Conducted exploratory data analysis using SQL and Python to identify trends and support business initiatives.
- Collaborated with cross-functional teams to design scalable data schemas for a new analytics platform.

Education

National University of Singapore Jan 2025 – June 2026
Master's in Computing – Artificial Intelligence Singapore

University of Waterloo Jan 2019 – Sep 2023
Bachelor's in Honours Mathematics, Statistics Major, Computing Minor Ontario, Canada

Projects

NUSAdvisor: AI Academic Advisor for NUS Students Project Link

- Built and deployed a chatbot using LangGraph and Gemini which allows NUS students to quickly search across 7,100+ modules and plan module selection using multi-turn dialogue, reducing hours of manual browsing and comparison on NUSMods.
- Extracted 7,100+ module and 160+ department records from the NUSMods API; embedded with all-MiniLM-L6-v2 (SentenceTransformers) and indexed in ChromaDB using HNSW with cosine similarity for fast semantic search.
- Utilized a router-tool architecture with 4 tools (semantic search, exact lookup, department resolution, course state tracking).
- Implemented persistent multi-session chat and streaming responses.
- Conducted evaluation-driven development using Arize Phoenix. Built custom code-based evaluators and LLM-as-judge evaluators. Achieved >85% score on all evaluators.
- Containerized with Docker and deployed on AWS using EC2, EBS for persistent storage, and S3 with CloudFront for CDN and HTTPS.

Bot-or-Not: Twitter Bot Detection Project Link

- Developed an end-to-end bot detection pipeline using Graph Neural Networks on TwiBot-22 dataset.
- Constructed a discrete-time dynamic graph (∼800 snapshots) to model evolving user behavior and interaction patterns.
- Engineered feature representations combining user metadata, profile signals, and RoBERTa-based text embeddings from tweets and descriptions.
- Implemented BotRGCN (static baseline) and EvoRGCN (temporal) models; achieved +1.87% macro-F1 improvement by capturing temporal behavioral drift.

Publications

- Deglint J. L., **Chandiramani K.**, Woo S., et al. "Investigation of Unsupervised Auto-segmentation for Weak Phytoplankton Annotations." *Journal of Computational Vision and Imaging Systems* – ***JCVIS 2022***

Certifications

AI Agents in LangGraph May 2026
DeepLearning.AI

Evaluating AI Agents May 2026
DeepLearning.AI